AI for Digital Characters

Lecture Slide Summaries (L01-L14)

L01 - Introduction

Course overview and digital character pipeline

Course goals

- Build conversational digital characters end-to-end
- Speech recognition, LLM responses, speech synthesis, animation
- Autonomous agents, personality, and emotions

Assessment

- Written exam: 120 minutes, 120 points, no summary sheet
- Non-programmable calculator and neutral dictionary allowed
- Optional quizzes in 7 lectures: 25/35 correct -> +0.25 grade bonus

Pipeline architecture

- User speech -> Speech Recognition -> transcript
- Transcript + emotion/intention/personality -> Chatbot (LLM or dialogue tree)
- Response text -> Speech Synthesis + Animation Synthesis
- Feedback loop: user emotion detected from text, speech, video

Course topics (lecture map)

- Affective Computing -> Deep Learning -> Speech Recognition
- LLMs (3 lectures) -> Speech Synthesis (2) -> Animation (2)
- Autonomous Agents -> Applications -> Reinforcement Learning

Use cases

- Digital Einstein: ETH branding, chatbot + TTS + animation
- Healthcare, education, museums, metaverse companions
- Location-based and mixed-reality experiences

Speech synthesis history (preview)

- Concatenative -> statistical parametric -> neural end-to-end
- Training costs of modern LLMs are substantial (industry context)

L02 - Affective Computing

Emotion theory, video features, and biosensors

What is affect?

- Broader than emotion: moods, attitudes, interpersonal stances
- Three ingredients of human behavior: cognition, affect, behavior
- Applications: education (learning gain), health, digital characters

Emotion theories

- James-Lange: Stimulus -> physiological response -> emotion (sequential)
- Cannon-Bard: physiological arousal and emotion occur simultaneously
- Schachter-Singer: arousal + cognitive interpretation -> emotion
- Dimensional: valence, arousal, dominance (Russell's core affect)
- Basic emotions (Ekman): joy, sadness, disgust, surprise, anger, fear

Video-based recognition

- Pipeline: preprocessing -> feature extraction -> classification (valence/arousal)
- Action Units (FACS): 46 AUs, independent facial muscle motions
- Examples: Happiness = AU6+12; Surprise = AU1+2+5+26
- Eye blinks: AU45 per frame; Eye gaze: 9 regions
- Mouth Aspect Ratio: MAR = (|p2-p8|+|p3-p7|+|p4-p6|) / (3*|p5-p1|)
- Fidgeting: frame diff -> threshold -> energy % -> background update
- Classifiers: Random Forest, SVM, TCNs, Transformers, CNNs on face images

Biosensors

- ECG: electrical heart activity from skin electrodes
- PPG: light reflected by blood vessels (also via webcam)
- HRV: std of R-R intervals, RMSSD, pNN50; higher HRV -> relaxed/adaptive
- Skin conductance (GSR): sweating level, indicative of arousal
- SC decomposition: phasic (event-related) + tonic (baseline) via convex optimization (cvxEDA)
- Phasic features: amplitude, latency, rise time, half-recovery time

Artifact detection (R-R intervals)

- QD = (Q3 - Q1) / 2
- MAD = (Median Beat - 2.9*QD) / 3
- MED = 3.32*QD
- CBD = (MAD + MED) / 2
- Beat is artifact if |RR - last_valid| > CBD

Ground truth

- Self-Assessment Manikin (SAM): valence, arousal, dominance
- Other sensors: EEG, EMG, respiration

L03 - Deep Learning Preliminaries

Neural nets, RNNs, attention, and Transformers

Neural network unit

- z = w*x + b; output a = sigma(z)
- Activations: ReLU (most common), sigmoid [0,1], tanh [-1,1]
- Softmax: converts logits to probabilities summing to 1
- Training: forward pass -> loss (cross-entropy) -> backpropagation

Feedforward networks

- Input -> hidden (ReLU) -> output (sigmoid or softmax)
- Applications: sentiment classification, language modeling

Recurrent Neural Networks

- h_t = sigma(U*h_{t-1} + W*x_t); y_t = f(V*h_t)
- Shared weights U, V, W across time steps; processed sequentially
- Captures sequential data via hidden state memory
- Vanishing gradient problem limits long-range dependencies
- LSTM: forget, input, output gates + cell state highway
- Bidirectional RNN: left-to-right + right-to-left, concatenate hidden states
- Stacked RNNs: multiple layers for richer representations
- Uses: language modeling, sentiment classification (last hidden state), generation
- Teacher forcing: use gold previous token during training

Encoder-decoder

- Encoder summarizes input (x1…xT) into context vector h_T
- Decoder generates output sequence autoregressively
- Applications: machine translation, summarization, dialogue
- p(y|x) = product of conditional next-token probabilities

Attention mechanism

- Problem: RNN bottleneck and vanishing gradients for long sequences
- Score: sim(q, k_t) = q*k_t / sqrt(d) [scaled dot-product]
- Weights: a_t = softmax(scores); Context: c = sum a_t * v_t
- Self-attention: Q, K, V from same sequence
- Cross-attention: Q from decoder, K/V from encoder
- Multi-head attention: multiple parallel attention heads

Transformers

- No inherent order -> positional encoding added to embeddings
- PE(t,2i) = sin(t/10000^(2i/d)); PE(t,2i+1) = cos(t/10000^(2i/d))
- Encoder: multi-head self-attention + FFN + skip connections + layer norm
- Decoder: masked self-attention (no future peeking) + cross-attention
- Layer norm: normalize across sequence per example (not batch norm)

L04 - Speech Recognition

Phonetics, features, ASR architectures, and WER

Phonetics

- Phones: speech sounds (ARPAbet/IPA symbols)
- F0 = lowest frequency of periodic waveform; pitch = perceptual correlate
- Mel scale: Mel = 1127*ln(1 + f/700) - perceptual pitch spacing
- Complex waves = sum of sine waves; spectrogram shows frequency over time
- Consonant types: fricatives, stops; vowels have formant structure

Feature extraction pipeline

- 1. Windowing: 20-40 ms frames, ~50% overlap, Hamming window
- 2. DFT/FFT per frame -> power spectrum P(k)
- 3. Mel filter bank: triangular overlapping filters on Mel scale
- 4. Log mel -> ~80-dimensional feature vector per frame
- Augmentation: time warping, frequency masking, time masking

Mel filter bank construction

- Convert f_low and f_high to Mel
- Space N+2 points linearly in Mel (for N filters)
- Convert back to Hz; create triangular filters (start-peak-end)

ASR architectures

- Encoder-decoder with subsampling (CNN pooling, pyramidal RNN)
- CTC: per-frame output + blank token; collapse duplicates; alignment via dynamic programming
- RNN-Transducer: encoder + predictor (LM) + joiner; supports streaming
- Streaming: increment output u on non-blank, acoustic index t on blank
- Whisper: Transformer encoder-decoder, multilingual multitask

Word Error Rate

- WER = 100 x (Substitutions + Insertions + Deletions) / N_ref
- Lower is better; function-word errors common
- Statistical significance: matched-pair sentence segment word error test

L05 - Chatbot I: Foundations

NLP, embeddings, generation, and LLM adaptation

Chatbot basics

- Definition: program simulating human conversation (AI not required)
- Rule-based: deterministic, precise, low risk; poor scalability/flexibility
- Retrieval vs generation: encoder-encoder (dot product) vs encoder-decoder

Text representation

- Tokenization: BPE sub-word tokens
- BoW -> tf-idf -> dense embeddings (Word2Vec, GloVe)
- Encoder models (BERT): bidirectional, good for understanding
- Decoder models (GPT): autoregressive, good for generation

Response generation

- Language model: $P(x_{t+1} | x_1, \ldots, x_t)$
- Greedy: always highest prob (repetitive)
- Top-k: sample from k most likely tokens
- Top-p (nucleus): smallest set with cumulative prob >= p
- Temperature: sharpens/flattens distribution
- Beam search: keep top-B partial hypotheses

LLM adaptation

- Fine-tuning: update weights on task data (needs compute + data)
- Prompt engineering: in-context examples, no weight updates
- Instruction tuning (FLAN/T5), MMLU multitask pretraining
- RLHF: reward model + policy optimization (InstructGPT/ChatGPT)
- Prompting: zero-shot, few-shot, chain-of-thought, self-ask, ReAct

Optimizations

- PEFT / LoRA: train low-rank adapters, freeze most weights
- Pruning: remove least important weights
- Knowledge distillation: teacher -> smaller student
- Quantization: lower precision (AWQ, SmoothQuant)

L06 - Chatbot II

Prompting, RAG, and chatbot evaluation

Prompting

- Roles: system (persona), user (input), assistant (history)
- In-context learning: examples at inference, no gradient updates
- Structured outputs: JSON schema, function calling / tool use
- ReAct: Thought -> Action -> Observation loop
- DSPy: automatic few-shot example selection

Retrieval Augmented Generation (RAG)

- Solves: hallucination, stale/private knowledge
- Pipeline: chunk documents -> embed -> retrieve -> augment prompt -> generate
- Chunking: fixed-size, recursive, semantic
- Sparse retrieval: tf-idf, BM25 + inverted index
- Dense retrieval: bi-encoder (SBERT), ColBERT (token MaxSim)
- Vector DBs: FAISS, Pinecone, Milvus, Weaviate (ANN search)
- Reranking: bi-encoder (fast, top-100) -> cross-encoder (precise, top-5)
- Evaluation: grounding score, answer relevance

Chatbot evaluation

- Qualitative: MOS (1-5 rating), A/B preference tests
- Quantitative: perplexity, BLEU, cosine similarity
- Perplexity: $P = 2^{(-1/N \cdot \sum \log_2 P(w_i))}$ - lower is better
- LLM-as-a-Judge: strong LLM rates responses
- Benchmarks: Chatbot Arena (Elo), MMLU, Stanford HELM

L07 - Chatbot III: Multimodality

Multimodal input, emotion recognition, personality

Full interaction pipeline

- Speech-to-text -> feature recognition -> user state -> response generation
- -> chatbot state -> speech synthesis + animation
- User state: emotion, attention, age, gender, appearance, etc.

Emotion recognition

- Taxonomies: VAD (dimensional) vs basic emotions (categorical)
- Visual: landmarks -> CNN (VGG, EfficientNet) or ViT/Swin
- Audio: pitch, rhythm, intonation, energy from Mel spectrogram
- FACS for facial action units
- Fusion: early (concatenate features), late (fuse predictions), hybrid

Multimodal models

- CLIP: contrastive text-image pre-training, shared embedding space
- VLMs: joint visual + text processing (e.g. LLaVA)
- Personalization: user-specific fine-tuning on labeled data

Personality modeling

- Human Big Five (OCEAN): openness, conscientiousness, extraversion, agreeableness, neuroticism
- Chatbot traits: artificiality, conscientiousness, vibrancy, civility, neuroticism
- Dynamic Personality Infusion: trait intensity (-2 to +2) rewrites LLM response

L08 - Speech Synthesis I

Text analysis, prosody, diphone and unit selection

TTS pipeline overview

- Stage 1: Text Analysis (normalization, phonetics, prosody)
- Stage 2: Waveform Synthesis (concatenative or parametric)

Text normalization

- Identify tokens -> chunk sentences -> classify types -> expand to words
- Handles: abbreviations, numbers, punctuation disambiguation
- Homographs: live/li:v/ vs /laiv/ via POS tagging + dictionary

Phonetic analysis

- CMUdict: ~127K words, 39 phonemes, stress markers 0/1/2
- G2P: ML classifier for out-of-vocabulary words
- Names: ~20% of tokens; morphology and rhyme rules

Prosodic analysis

- Prominence/accent: which syllable is stressed in context
- Boundaries: intonation phrase (||), intermediate phrase (|)
- Duration and F0 (fundamental frequency) per phone
- Stress = lexical (fixed in word); accent = context-dependent
- Prominence classes: unaccented, reduced, pitch accent, nuclear accent
- ToBI: H* (peak), L* (low), L-H% (continuation rise)

Diphone synthesis

- Diphone: half of each adjacent phone - captures coarticulation
- Record -> segment -> concatenate with Hanning window -> prosody modify
- Epoch labeling: vocal fold cycle markers (Praat)

Unit selection synthesis

- Larger units from ~10h recorded speech; minimal signal processing
- Target cost T(s_t, u_j): match F0, stress, duration, position
- Join cost J(u_j, u_{j+1}): smoothness at boundaries (=0 if consecutive in DB)
- Select sequence minimizing sum of target + join costs

L09 - Speech Synthesis II

Acoustic models, vocoders, style transfer

Modern neural TTS

- Text analysis -> Acoustic model (mel spectrogram) -> Vocoder (waveform)

Tacotron 2

- Encoder: char embedding + conv layers + bi-LSTM
- Autoregressive decoder with attention -> mel frames
- Postnet refines mel spectrogram
- Pros: high quality; Cons: slow, attention instability (skip/repeat)

FastSpeech / FastSpeech 2

- Non-autoregressive: duration predictor + length regulator
- Parallel decoding -> fast inference, stable output
- FastSpeech 2: variance predictors for duration, energy, pitch
- Speed control: scale duration D (2xD -> half speed)
- Uni-TTS: multilingual FastSpeech 2 base (Microsoft Azure)

Vocoders

- Griffin-Lim: iterative phase from magnitude, no training, robotic
- WaveNet: autoregressive dilated causal convolutions, high quality, slow
- HiFi-GAN: GAN with multi-scale + multi-period discriminators, real-time
- DiffWave: diffusion-based, high quality, slow inference
- HiFi-GAN losses: adversarial + feature matching + L1 mel loss

Style / speaker control

- One-hot speaker labels vs learned speaker embeddings
- Embeddings enable interpolation and zero-shot adaptation
- Style tokens: unsupervised style discovery

Evaluation

- MOS: 1-5 naturalness rating; A/B preference tests
- DRT: 96 rhyming pairs in carrier phrases
- MRT: 50 sets of 6 similar words
- WER via ASR on synthesized speech (intelligibility)

L10 - Animation I

Principles, anatomy, kinematics, and skinning

Character fundamentals

- Stylized vs realistic: stylized avoids uncanny valley
- Uncanny valley: appeal drops as human-likeness increases (before real human)

Disney's 12 principles (know for exam)

- 1. Squash and Stretch - mass and rigidity
- 2. Anticipation - preparation before main action
- 3. Staging - clear focus
- 4. Straight Ahead vs Pose to Pose
- 5. Follow Through and Overlapping Action
- 6. Slow In and Slow Out - ease in/out (NOT linear)
- 7. Arc - natural curved paths
- 8. Secondary Action
- 9. Timing
- 10. Exaggeration
- 11. Solid Drawing
- 12. Appeal

Anatomy and rigging

- Facial muscles: bone-to-skin; wrinkles perpendicular to contraction
- Joints: hinge, ball-and-socket, etc.; 0-6 degrees of freedom
- Skeleton hierarchy: bones define movement, joints define rotation

Kinematics

- Forward kinematics (FK): joint angles -> end-effector position
- Inverse kinematics (IK): target position -> joint angles
- Jacobian method: Deltatheta = alpha * J+ * Deltae
- Pole vector: controls elbow/knee bend direction in IK
- Mass-spring systems for secondary motion

Skinning

- Linear Blend Skinning (LBS): v' = sum w_ij * (R_j * v_i + T_j)
- Each vertex influenced by multiple bones with weights summing to 1
- Candy-wrapper artifact: unnatural twisting at large joint angles
- Motion capture: blend weights per vertex

L11 - Animation II

Data-driven animation and speech-driven faces

Why ML for animation?

- MoCap cannot cover all possible chatbot responses
- Procedural methods are repetitive; need on-the-fly synthesis
- Active listening: generate plausible non-speaker animations with context

Datasets

- BIWI 3D, MEAD, VOCASET: 3D face scans with emotion
- IEMOCAP: dyadic dialogs, multi-modal, 9 emotions + VAD
- BEAT, Talking with Hands: co-speech gestures

Learning-based methods

- Holden's methods: phase-functioned neural networks for locomotion
- DeepPhase: phase-aware autoencoder for periodic motion manifolds
- Normalizing flows: invertible maps, exact likelihood via Jacobian determinant
- Applications: diverse motion generation with temporal consistency

Speech-driven animation

- Procedural: amplitude from audio waveform
- FaceFormer: transformer for speech-driven 3D facial animation
- Blendshapes: brow, eyes, jaw, mouth controls

Evaluation

- FID: realism of generated motion distribution
- Foot sliding / contact error for locomotion

L12 - Autonomous Agents

Agent properties, dialogue trees, and knowledge graphs

Agent definition and properties

- Agent: perceives environment via sensors, acts via actuators
- Reactivity: timely response to environmental changes
- Autonomy: operates without direct human control
- Proactiveness: plans for future opportunities (not just reacting)
- Socially intelligent agents: real-time perception, emotion, action in social settings

Decision-making approaches

- Finite State Machines: states + transition conditions
- Dialogue trees: hierarchical choices; low flexibility, hard to scale
- LLM-based: task decomposition, tool use, continuous interaction

Knowledge graphs

- RDF triples: (subject, predicate, object)
- Schema design with typed entities and relations
- SPARQL: W3C standard query language for structured retrieval
- Query types: one-hop, path, conjunctive

Knowledge graph embeddings

- TransE: $h + r \sim t$; score = $||h + r - t||$
- TransE limitation: symmetric relations ($r \sim 0$)
- TransR: entities and relations in different spaces
- Link prediction: infer missing edges in large/incomplete graphs
- Query2box: box embeddings for complex conjunctive queries

LLM + KG integration

- KG grounds LLM facts; reduces hallucination
- Use cases: smart home coordination, healthcare, recommendations, bias mitigation

L13 - Metrics & Applications

Evaluation, ethics, and case studies

TTS evaluation (recap)

- MOS, A/B tests, DRT, MRT, WER for intelligibility

Responsible AI

- Harmful system: negative real-world impact
- Biased system: different performance across subpopulations
- Bias sources: training data, system design, distribution shift
- ASR racial disparities; face recognition demographic bias
- Problem formulation should be ethical (e.g. predict criminality from face = bad task)

Case studies

- Digital Einstein: image generation, upcoming leg/feet animation
- Virtual Doctor: experimental user study setup
- Virtual Psychotherapist: human vs AI therapist evaluation

Exam preparation

- Review all pipeline components and evaluation metrics
- Know trade-offs: rule-based vs LLM, concatenative vs neural TTS, etc.

L14 - Reinforcement Learning

MDPs, policy optimization, and behavior synthesis

ML paradigms

- Supervised: (data, label) pairs, backpropagation
- Unsupervised: learn latent representations (autoencoders)
- RL: agent-environment interaction, reward signal (no labels)
- RL for non-differentiable simulators (physics, game engines)

MDP and key concepts

- State s, action a, reward r, policy pi(a|s), return R (discounted sum)
- Trajectory τ: sequence of (s, a, r) from one episode
- Markov property: future depends only on current state

Value functions

- Q(s,a): expected return from taking action a in state s
- V(s): E_{a~pi}[Q(s,a)]
- Optimal policy maximizes expected return (not each individual reward)

Algorithms

- Policy gradient: optimize pi directly via sampled trajectories
- Actor-Critic: critic baseline reduces variance
- PPO: clipped surrogate objective, on-policy rollouts
- Q-learning: implicit policy = argmax_a Q(s,a); off-policy with replay buffer
- Reward does NOT need to be differentiable

PPO advantages

- Critic baseline reduces variance
- Clipped surrogate prevents destructively large policy updates
- Multiple epochs on same batch improves sample efficiency

Behavior synthesis applications

- Humanoid locomotion and goal reaching
- Soccer dribbling, motion tracking in simulation
- Formulate: state, action, reward, policy for each task
- Prior knowledge (e.g. z ~ N(0,I)) can be encoded as KL penalty